

beneficial for those new to it to quickly access an online tool rather than navigate the challenges of installing it on their preferred operating system. However, it is now time to explore alternative tools and options beyond CoCalc.

## Installing and using SageMath on your computer

Let us start by installing SageMath on your computer. The installation process can vary depending on the version of SageMath you choose and the operating system you use. As mentioned earlier in this series, the latest version of SageMath supported by CoCalc is version 10.4. Installing this version manually on your computer can be a lengthy process, but you can find detailed instructions on the SageMath website.

If you are open to using a slightly older version, the process becomes much easier. For instance, installing SageMath on Ubuntu is straightforward. Simply open the GNOME Software application (commonly referred to as Ubuntu Software or Ubuntu Software Center), search for 'SageMath', and click the *Install* button. Enter your root password when prompted, and the installation will be completed in no time. Other Linux distributions and operating systems likely offer a similarly simple process. However, note that this method typically installs version 9.0 of SageMath.

I have been using version 9.0 for some time and find it perfectly capable. Personally, I prefer sticking with stable software rather than frequently updating just to gain a feature I may never use. Ultimately, the choice of version and installation method is yours, based on your needs.

Assuming you have a reasonably recent version of SageMath installed, let us explore how to use it locally. SageMath is like a curious and resourceful individual—it adopts the best tools it encounters. For example, SageMath integrates the IPython shell into its interactive environment. This enhanced IPython-based interface offers powerful features like magic commands (`%time`, `%timeit`, `%run`, etc), tab completion, and syntax highlighting.

To experience this, open a terminal, type the command `sage`, and press the *Enter* button. This will launch the SageMath interactive shell (Figure 1).

As shown in Figure 1, I am using SageMath version 9.0 along with Python 3.8.10. I subscribe to the philosophy, "If it ain't broke, don't fix it." I only update software when it directly impacts my work, rather than simply because a new version is available and I have received an update notification.

In the example commands shown in Figure 1, the first two lines demonstrate execution of basic mathematical operations and display the results. The third and fourth lines illustrate how to measure the execution time of calculations using the `%timeit` magic command. This command executes the expressions `'0.99^100000'` and `'1.1^100000'` multiple times, calculating an average execution time. Such an approach provides a more reliable performance assessment, particularly for operations that execute very quickly. From Figure 1, it is evident that SageMath is both fast and powerful, making it an invaluable computational tool. Finally, the `exit()` command is used to exit the shell, functioning similarly to its usage in IPython.

Now, let us learn how to execute saved SageMath programs. Consider the single-line SageMath program named `test.sage`, as shown below.

```
print("This is a very simple SageMath code")
```

Note that this is essentially Python code in disguise, as SageMath is built on Python syntax. To run the program `test.sage`, execute the following command in a terminal:

```
sage test.sage
```

You will now see the text "This is a very simple SageMath code" printed on the terminal. But what happened to the `.sage` extension that we used for all the programs discussed

```
# sage

SageMath version 9.0, Release Date: 2020-01-01
Using Python 3.8.10. Type "help()" for help.

sage: 0.99^100
0.366032341273229
sage: 1.1^100
13780.6123398224
sage: %timeit 0.99^100000
1.7 µs ± 14.3 ns per loop (mean ± std. dev. of 7 runs, 1,000,000 loops each)
sage: %timeit 1.1^100000
1.69 µs ± 10.2 ns per loop (mean ± std. dev. of 7 runs, 1,000,000 loops each)
sage: exit()
Exiting Sage (CPU time 0m28.37s, Wall time 7m6.82s).
```

Figure 1: SageMath interactive shell